1. Procedure

1. Create the Experiment_10 project directory and initialize a Git repository.
2. Create the Python application file app.py and test the application locally.
3. Create requirements.txt containing the security tools Bandit and pip-audit.
4. Create the pipeline.sh script to automate repository verification, dependency installation, security scanning, and application testing.
5. Make pipeline.sh executable using chmod +x pipeline.sh.
6. Execute the CI/CD pipeline using ./pipeline.sh.
7. The pipeline checks the Git repository and installs the required security tools.
8. Bandit performs static security analysis of the Python source code.
9. pip-audit checks installed Python dependencies for known vulnerabilities.
10. The Python application is executed as the final test stage.
11. The pipeline displays the status of each stage and generates a final CI/CD pipeline result.

2. Source Code

2.1 app.py

```
def add(a, b):
    return a + b


def main():
    print("SSD CI/CD Security Demo")
    print("Result:", add(10, 20))


if __name__ == "__main__":
    main()
```

2.2 requirements.txt

```
bandit
pip-audit
```

2.3 pipeline.sh

```
#!/bin/bash

echo "================================================"
echo "       CI/CD PIPELINE WITH SECURITY TOOLS"
echo "================================================"

echo "[1/6] Checking Git Repository..."

if git rev-parse --is-inside-work-tree > /dev/null 2>&1; then
    echo "[OK] Git repository verified."
else
    echo "[ERROR] Git repository not found."
    exit 1
fi


echo
echo "[2/6] Installing Dependencies..."

pip3 install -r requirements.txt -q

echo "[OK] Dependencies installed successfully."


echo
echo "[3/6] Running Bandit Security Scan..."

bandit -r app.py
```

```
if [ $? -eq 0 ]; then
    echo "[OK] Bandit scan completed."
else
    echo "[WARNING] Bandit found security issues."
fi



echo
echo "[4/6] Running pip-audit Dependency Scan..."

pip-audit

if [ $? -eq 0 ]; then
    echo "[OK] No vulnerable dependencies found."
else
    echo "[WARNING] Dependency vulnerabilities detected."
fi



echo
echo "[5/6] Running Application Tests..."

python3 app.py

if [ $? -eq 0 ]; then
    echo "[OK] Application executed successfully."
else
    echo "[ERROR] Application test failed."
    exit 1
fi



echo
echo "[6/6] Generating Pipeline Report..."

echo "[OK] Git repository verified"
echo "[OK] Dependencies installed"
echo "[OK] Bandit security scan completed"
echo "[OK] pip-audit scan completed"
echo "[OK] Application tested successfully"

echo
echo "================================================="
echo "      CI/CD PIPELINE PASSED SUCCESSFULLY"
echo "================================================="
```

3. Output

